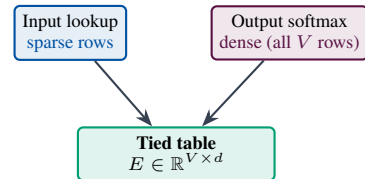


1. The problem: One tensor plays two roles



two structurally different gradients
merged on one tensor: $G_E = G_{\text{in}} + G_{\text{out}}$

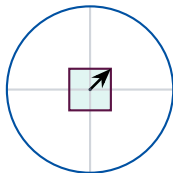
2. One first-moment LMO per parameter class — AdamW-free



3. Tied table cap sweep (2D visualization)

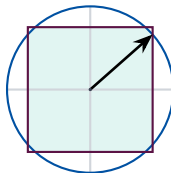
- Row- ℓ_2 ball ($\sqrt{d}$, fixed)
- ◻ Max-entry cap (half-width c)
- Shared intersection
- LMO update

Sign endpoint
 $c = 1$



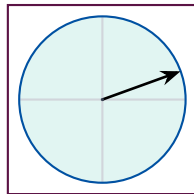
u_1^*

AF-Muon finite cap
 $1 < c < \sqrt{d}$



u_c^*

Row-RMS endpoint
 $c \geq \sqrt{d}$



u_{rms}^*

